Networks Decide for Us:

When networks shape behavior more than intentions

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Motivation

Collective behaviors

HOME SEARCH

The New York Times

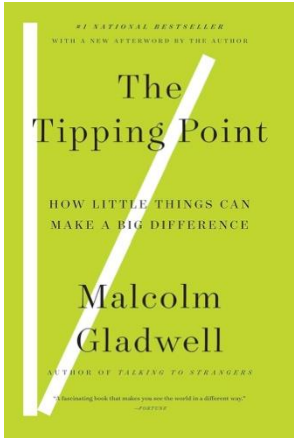
of Rare Earths Has U.S. Find Alternatives	U.S. Expands Antidrug Campaign With Strike on Boat in Pacific Near Colombia	Former Biden and Senate Counsel to Lead Progressive Legal Group	Maine U.S. Senate Candidate Says He Covered Up Tattoo That Had Nazi Imagery	The NATO chief and Trump will discuss support for Ukraine.	Nancy Pelosi Hasn't 2026 Plans. Scott Wi Anyway.

Crowd Scientists Say Women's March in Washington Had 3 Times as Many People as Trump's Inauguration

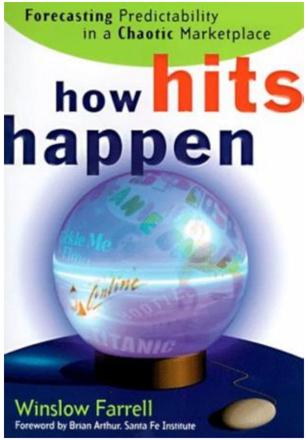
By TIM WALLACE and ALICIA PARLAPIANO UPDATED JAN. 22, 2017

Collective behaviors

1. Hush Puppies rapid growth in 95’
2. East New York and Brownsville crime reduction
3. Abrupt growth of fax users
4. Book: Divine Secrets of the Ya-Ya Sisterhood



[1]



[2]

1. Movies: The Full Monty
2. Market Share: AT&T
3. Literature: Hootie & the Blowfish

How are those collective behaviors studied?

Two main pathways of research - Collective Actions [3]

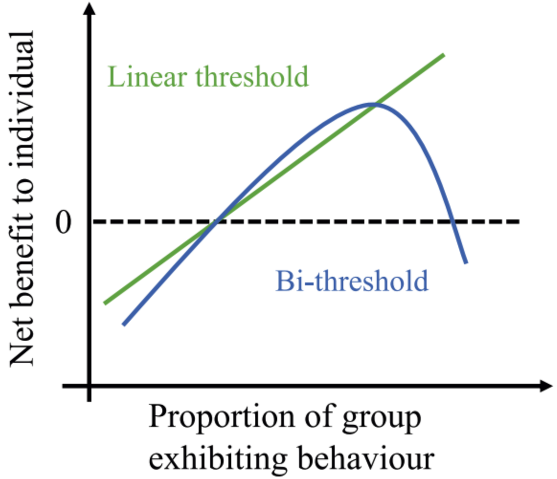
Collective action: *“any action which provides a collective good.”*

Granovetter’s **Threshold Model** (78’)**:** an individual's willingness to act depends on the number of others who are already participating.

Two main pathways of research - Collective Actions [3]

Threshold Models [4] :

*"The **threshold** is simply that point where the perceived benefits to an individual of doing the thing in question exceed the perceived costs."*



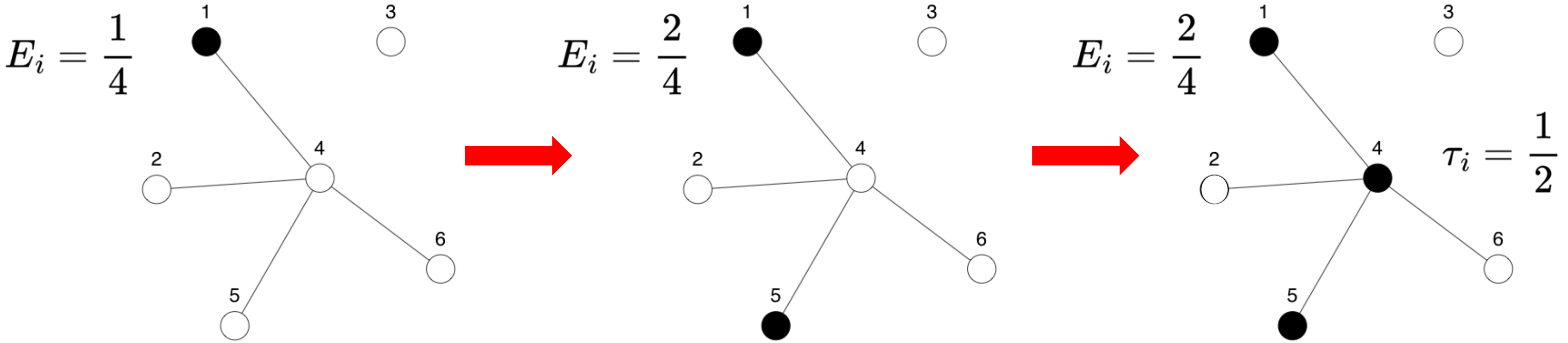
Thresholds denote the point where utility is positive, **not preferences**.

Two main pathways of research

Diffusion of Innovations [4] : *"a theoretical approach used to explain how new ideas and practices spread within and between communities."*

Valente's **Network Threshold Model** [5] :

$$E_i \equiv \frac{\sum_{j \neq i} \mathbf{X}_{ij} a_j}{\sum_{j \neq i} \mathbf{X}_{ij}}, \qquad a_i = \begin{cases} 1 & \text{if } \tau_i \leq E_i \\ 0 & \text{otherwise} \end{cases}$$



Two main pathways of research

Table 10–2. Event History Analysis of Factors Associated with Adoption for the Three Diffusion Network Datasets

	Medical Innovation	Brazilian Farmers	Korean Family Planning
No.	947	10,092	7,103
Time	132.7**	5.14	4.34
Cumulative adoption	0.03	2.27	0.32
Number sent	1.04	0.99	1.02
Number received	1.05	1.01	1.05**
Cohesion exposure	1.05	1.93**	2.08**
Structural equivalence exposure	1.04	5.01**	1.34
Science orientation	0.65**		
Journal subscriptions	1.67		
Cosmopolitan-ness		1.00	
Income		1.14**	
Number of children			1.23**
Media exposure			1.03*

Coefficients are adjusted odds ratios for likelihood of adoption. Estimates adjusted for clustering within community.

*$p < .05$; **$p < .01$.

Previous works

Others [4, 5] use *Percolation Theory* **instead** to construct a word-of-mouth diffusion model.

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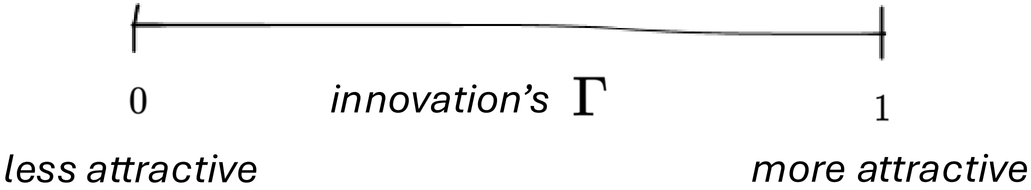

Previous works

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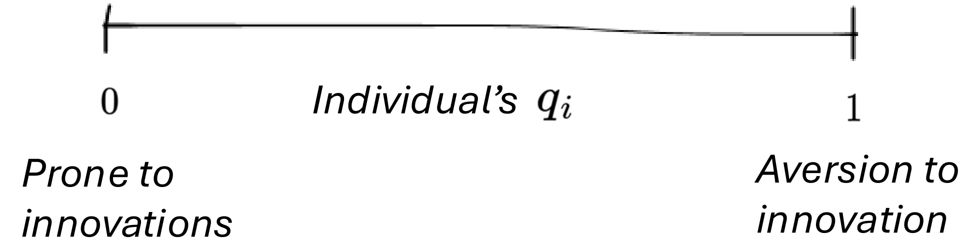
- An innovation has an *Intrinsic Utility Level* (IUL) $\Gamma \in [0,1]$

- Individuals are entirely characterized by
  - 1) their network position
  - 2) an attribute of *Minimum Utility Requirement* (MUR) $q_t^i \in [0,1]$



Preferences!

Previous works

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- An innovation has an *Intrinsic Utility Level* (IUL) $\Gamma \in [0, 1]$
- Individuals are entirely characterized by
  - 1) their network position
  - 2) an attribute of *Minimum Utility Requirement* (MUR) $q_t^i \in [0, 1]$
- There is an influence driven by adopters

$$q_t^i = q_0^i \left(\frac{1}{\sum_{j \neq i} \mathbf{X}_{ij} a_j} \right)^{\alpha}, \quad \alpha > 0$$

- Finally adopt if $q_t^i \leq \Gamma$.

Addressing the flaws

We want to see the effect of word-of-mouth diffusion with:

1. Plausible network topologies
2. The right MUR distribution
3. Tie-specific influence parameter $\alpha = \alpha_{ij}$

The model

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The model

1. Let's assume an 'innovation' has an *Intrinsic Utility Level* (IUL) $\Gamma \in [0,1]$, which characterize the attractiveness of that innovation.

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0 *innovation's* Γ 1

less attractive *more attractive*

2. And individuals with *Minimum Utility Requirement* (MUR) $q_i \in [0,1]$.

0 *Individual's* q_i 1

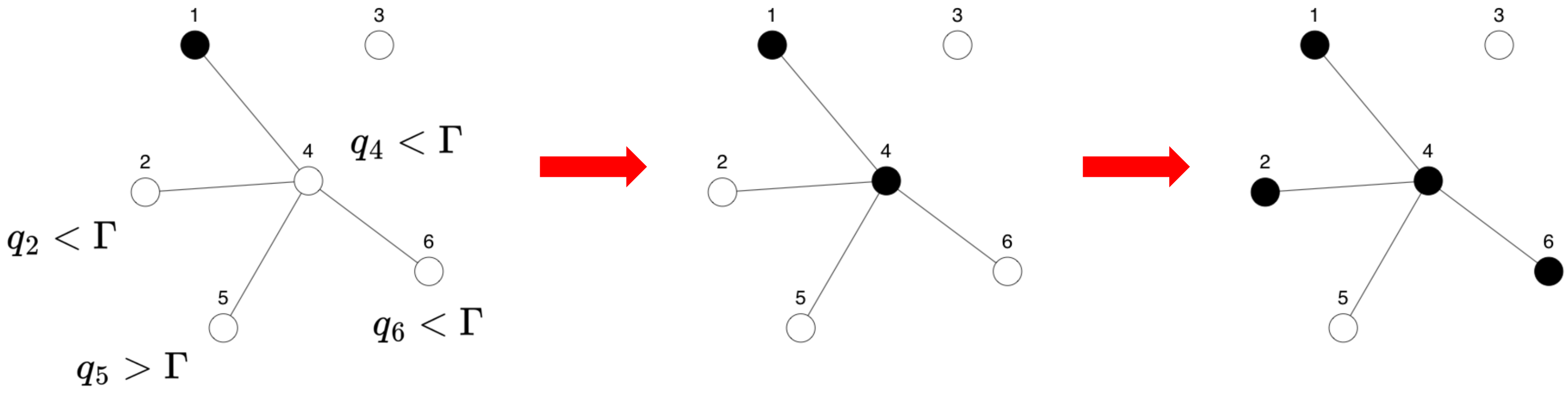
Prone to innovations *Aversion to innovation*

The model

Adoption can happen in two ways:

1) <u>Rational Choice:</u> i adopt if *Minimum Utility Requirement* ≤ *Intrinsic Utility Level*

$$q_i \leq \Gamma \qquad \Rightarrow a_i = 1$$



The model

2) <u>Selective Social Influence:</u>

$$E_i \equiv \frac{\sum_{j \neq i} \mathbf{X}_{ij} a_j}{\sum_{j \neq i} \mathbf{X}_{ij}}, \quad a_i = \begin{cases} 1 & \text{if } \tau_i \leq E_i \\ 0 & \text{otherwise} \end{cases}$$

The model

2) <u>Selective Social Influence:</u>

$$\tilde{E}_i \equiv \frac{\sum_{j\neq i} \mathbf{X}_{ij}\tilde{a}_j}{\sum_{j\neq i} \mathbf{X}_{ij}}, \qquad a_i = \begin{cases} 1 & \text{if } \tau_i \leq \tilde{E}_i \\ 0 & \text{otherwise} \end{cases}$$

Here, $\tilde{a}_i$ accounts for those infected individuals who *are influential*:

$$\tilde{a}_i = 1 \Leftrightarrow a_i = 1 \wedge U_{ij} \leq \sigma\left(h - d_{ij}\right), \quad U_{ij} \sim \mathrm{Unif}(0,1).$$

where:

1. d_{ij} is the social distance between individuals i and j,
2. h is the *Maximum Social Proximity* (MSP), which measures the flexibility to be influenced by a person with different demographics.

The model

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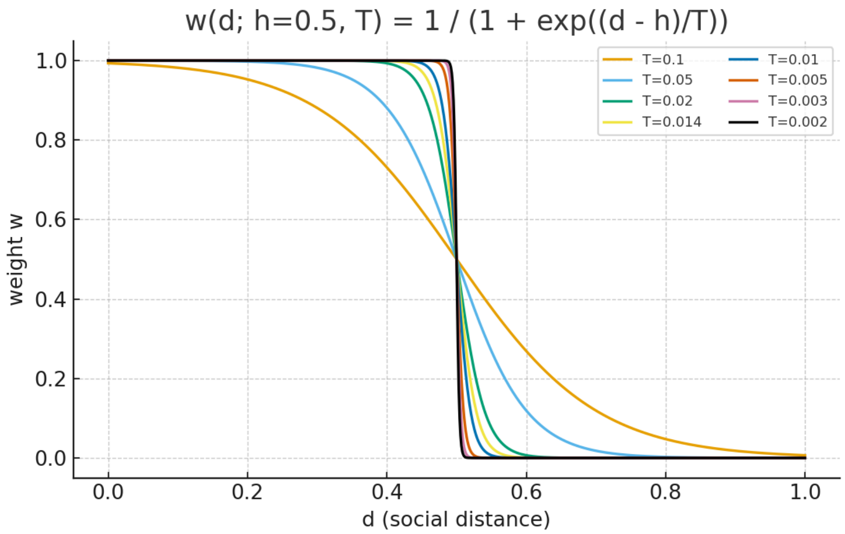
Where:

$$\sigma(z) = \frac{1}{1 + \exp(-z/T)}$$



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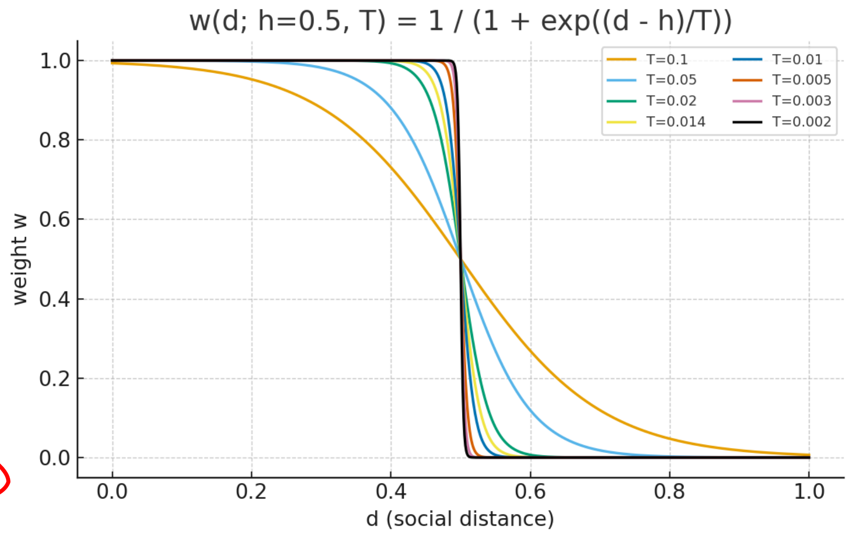
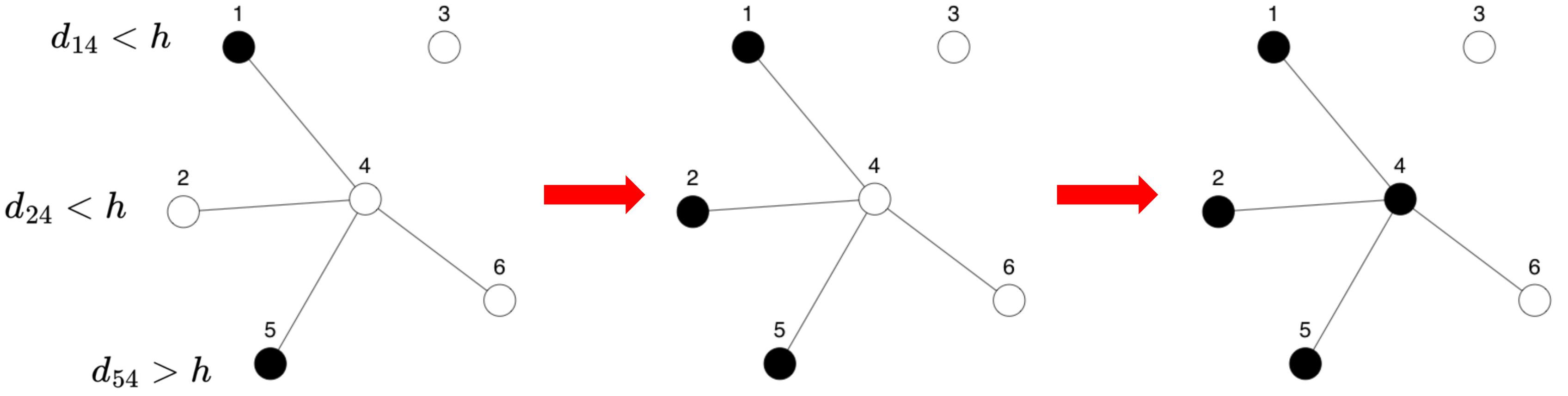
3. Tie-specific influence parameter ✓



The model

2) <u>Selective Social Influence:</u>

Let's set $\tau_4 = 0.5$; then, because the variability among the ties [6]...



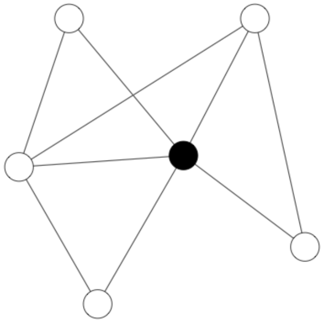
Setting up
–
Diffusion of Innovations

1. Plausible network topologies
2. The right MUR distribution

Imputing network structure to a Survey

Following McPherson and Smith (2019) [7], **you can impute a network structure** on any other survey if:

1) There is a survey with *representative* **EGO-data**,
2) **Both** are *representative* of the same population,
3) **Both** have some basic *demographic* variables.



Imputing network structure to a Survey

General Social Survey (GSS 2004):

1. Representative of the US population
2. Demographics:
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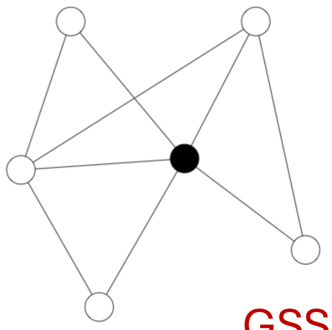

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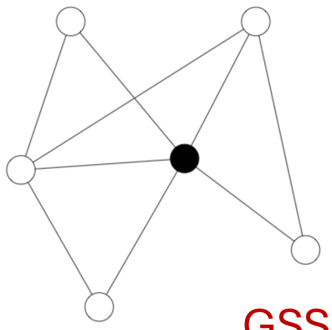

GSS 2004

American Trends Panel (ATP 2014):

1. Representative of the US population
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~~3. EGO data~~ '*Openness to Innovations*' items

	id_1	id_2	id_3	...
Age	36	27	41	...
Educ	12	15	16	...
q_i	.54	.67	.32	...
...	...	...	...	...

ATP 2014

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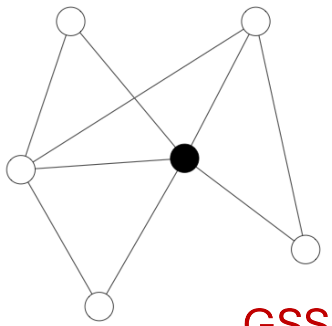

GSS 2004

Homophilic strength
in the m-th
social dimension

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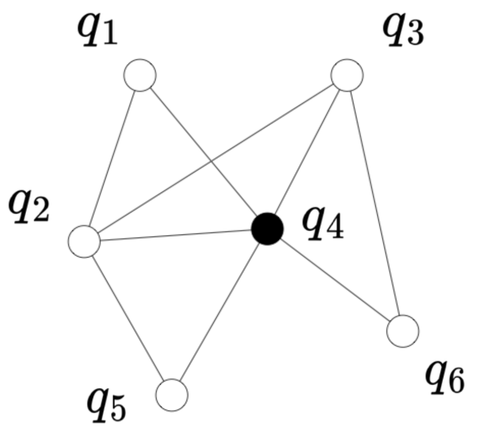

ATP 2014 +
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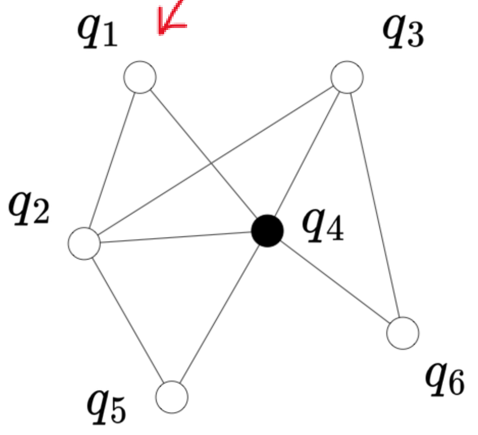
Nodes have ‘*Openness to Innovations* ***Score***’

MRU



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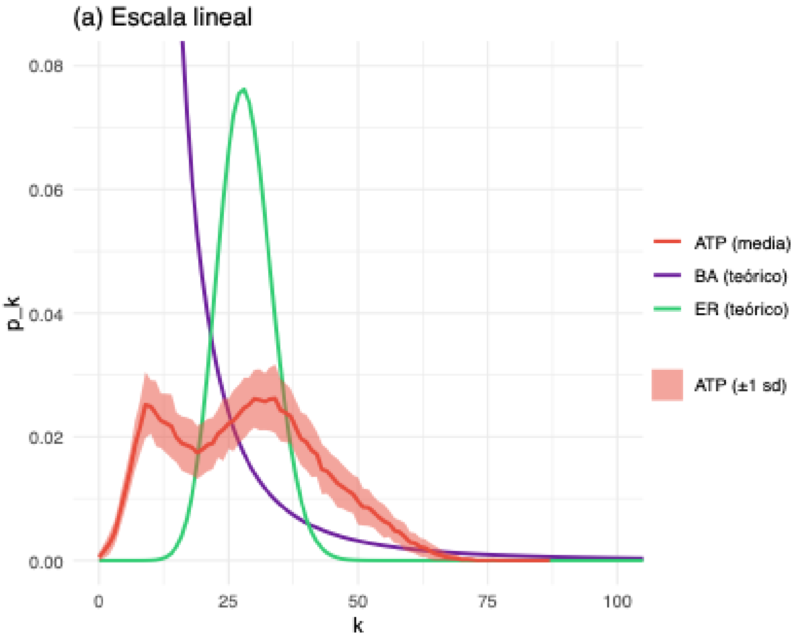

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✓✓

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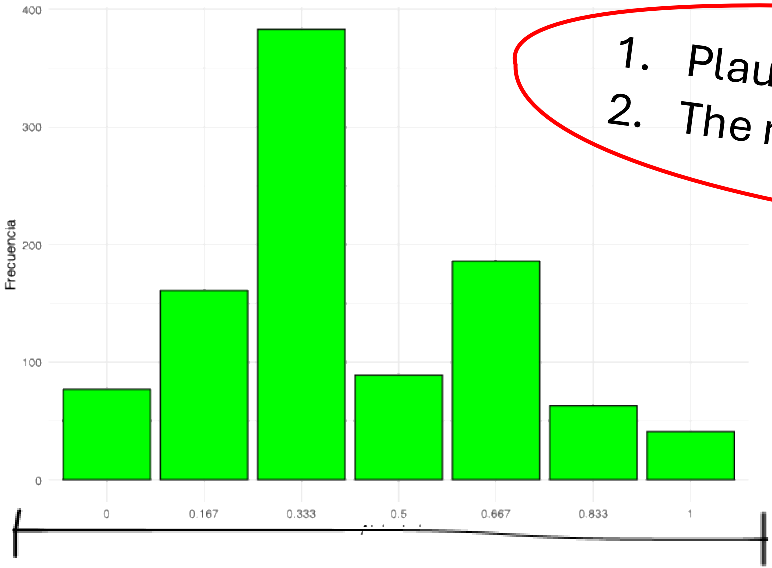
Prone to innovations



0

Individual's q_i

1

Aversion to innovation

Comprehensive Simulation Methodology

Using the ATP-networks (N=1000 nodes):

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Parameter Space Exploration:

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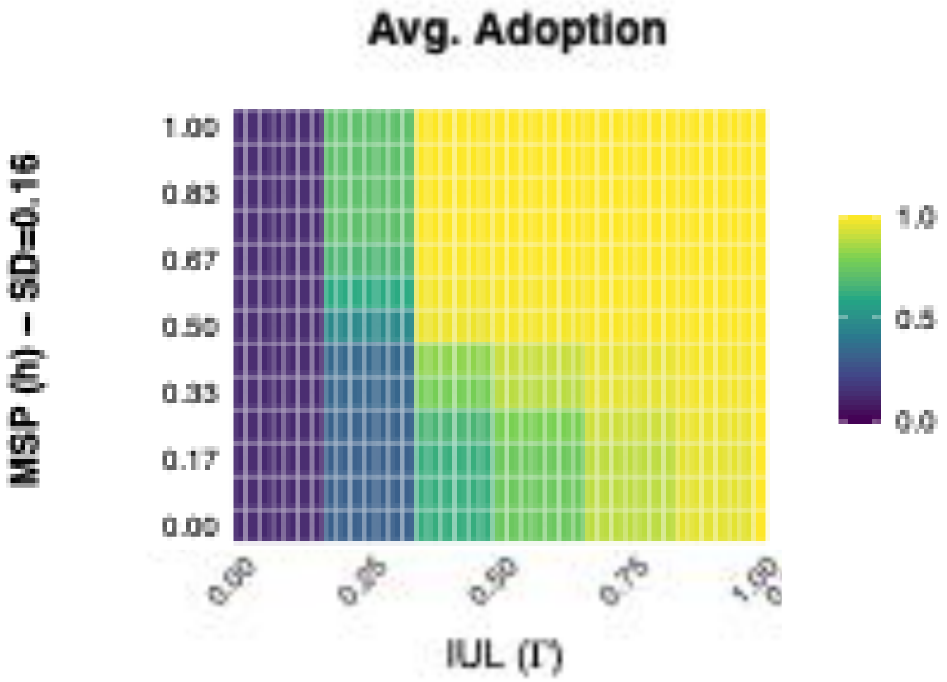
Scale of Simulation:

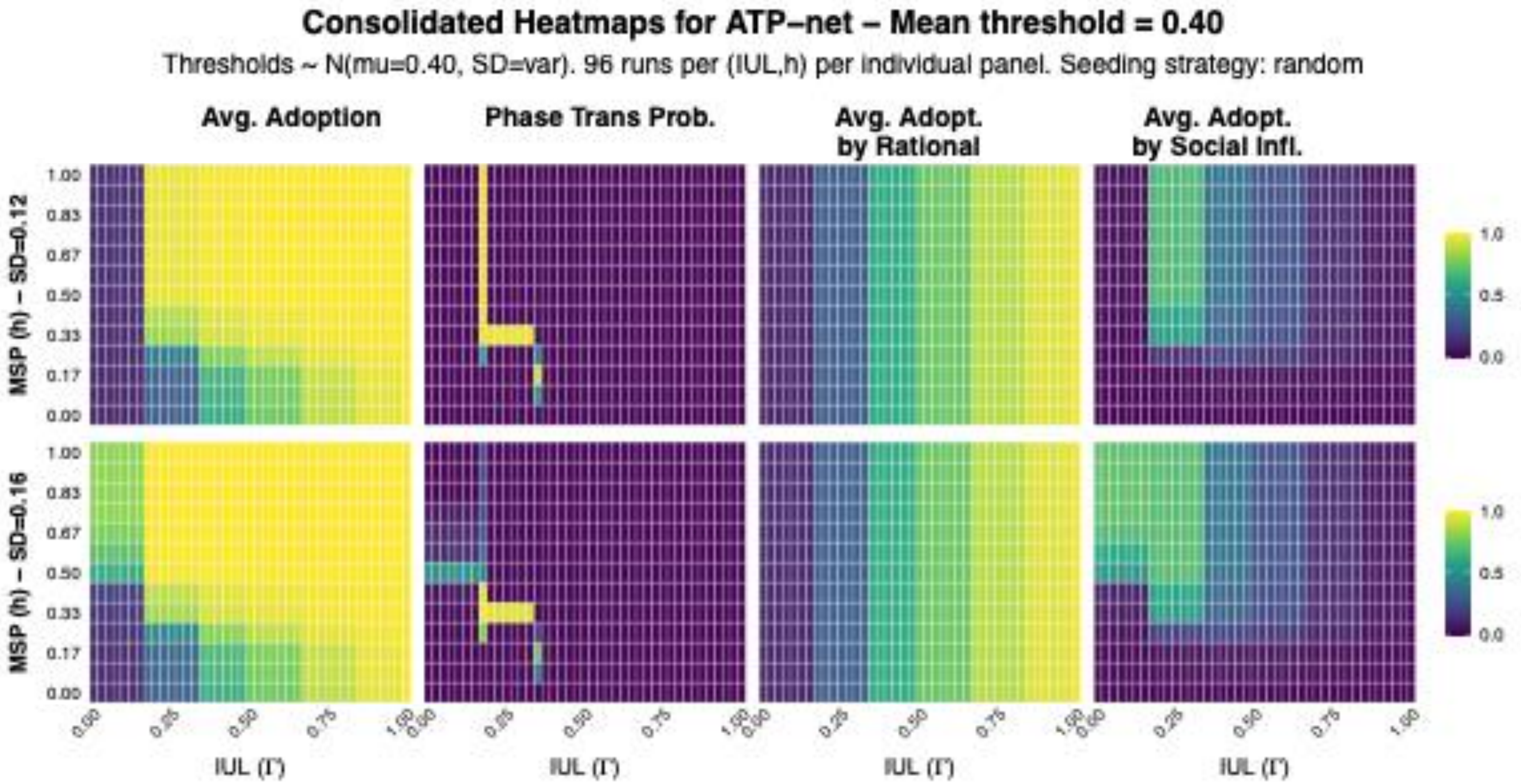
- 24 simulations for each combination of parameters
- Total simulations: $41 \times 13 \times 4 \times 4 \times 5 \times 24 \approx \underline{10^6}$!

Results
-
Diffusion of Innovation

Results


Avg. Adoption
MSP (h) – SD=0.16
1.00
0.83
0.67
0.50
0.33
0.17
0.00
0.00
0.25
0.50
0.75
1.00
IUL (Γ)
1.0
0.5
0.0

Results



Consolidated Heatmaps for ATP–net – Mean threshold = 0.40
Thresholds ~ N(mu=0.40, SD=var). 96 runs per (IUL,h) per individual panel. Seeding strategy: random
Avg. Adoption
Phase Trans Prob.
Avg. Adopt. by Rational
Avg. Adopt. by Social Infl.
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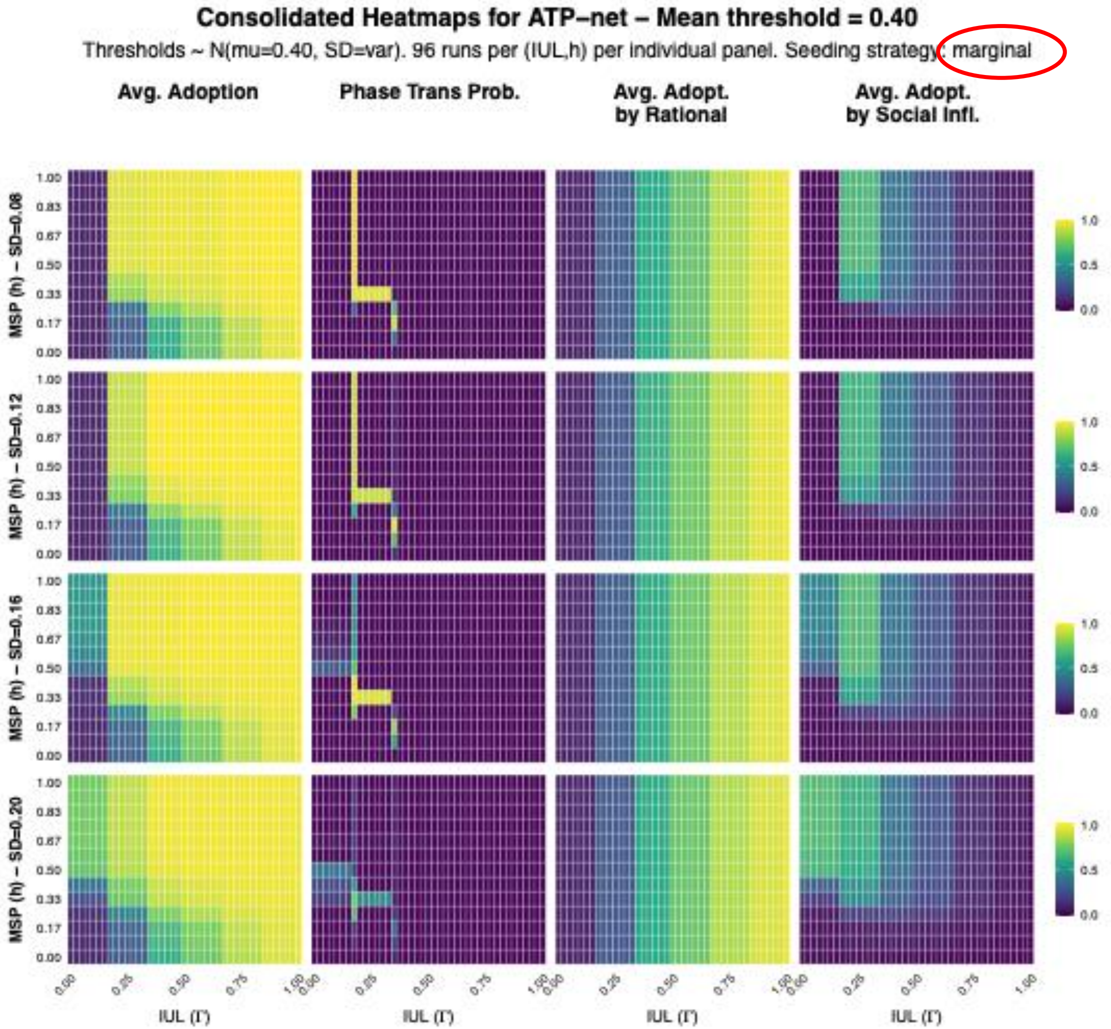

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Avg. Adoption
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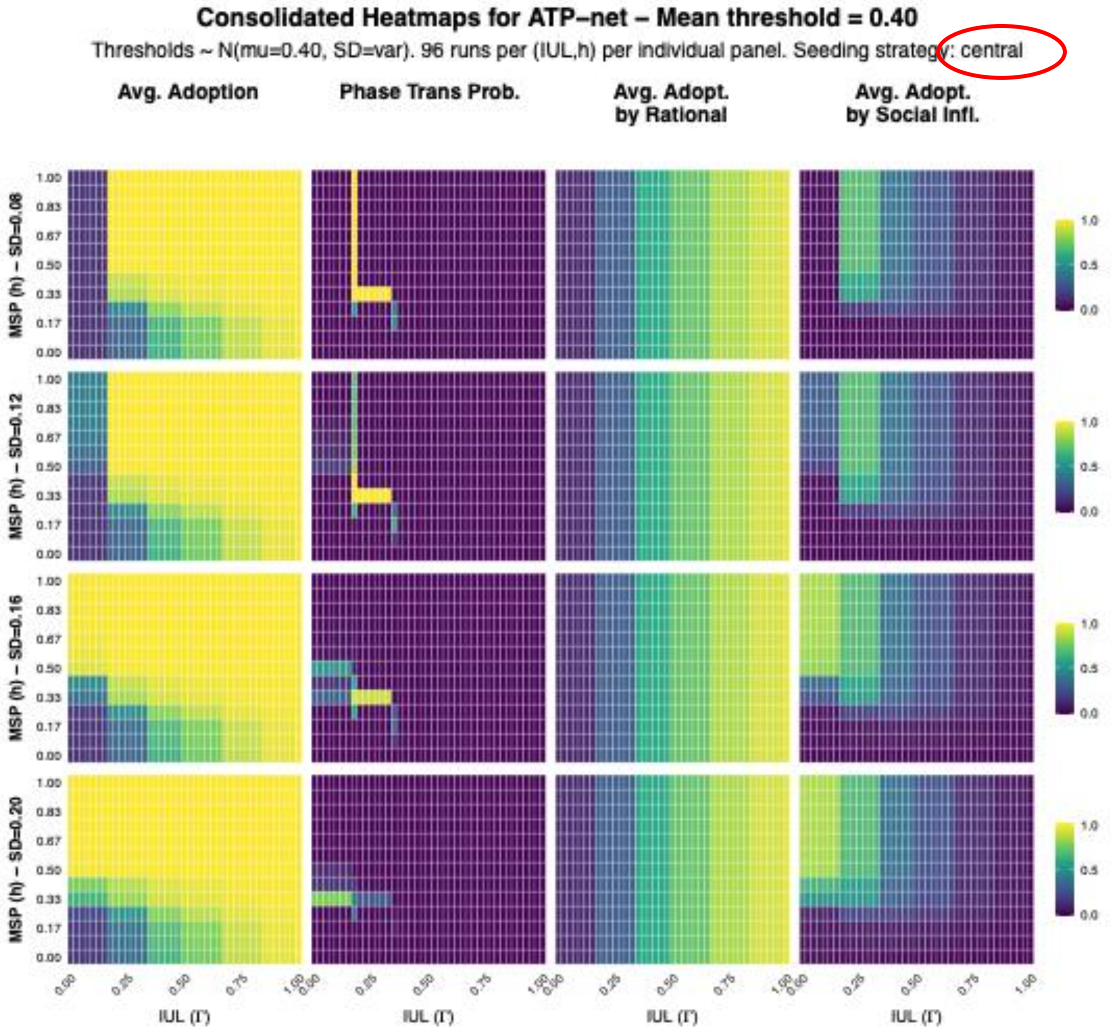
IUL (Γ)



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1.0
0.5
0.0

Conclusions – 1st part

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What have we learned?

- Social influence plays a relevant role in adoption, even for configurations where the innovation is **not particularly attractive**.
- The model offers consistent results across several parameter configurations.
- The regimens where social influence is on are separated abruptly by **first-order phase transitions**.
- There are non-structural barriers to word-of-mouth diffusion.

Setting up
–
Collective Action

1. Plausible network topologies
2. The right MUR distribution

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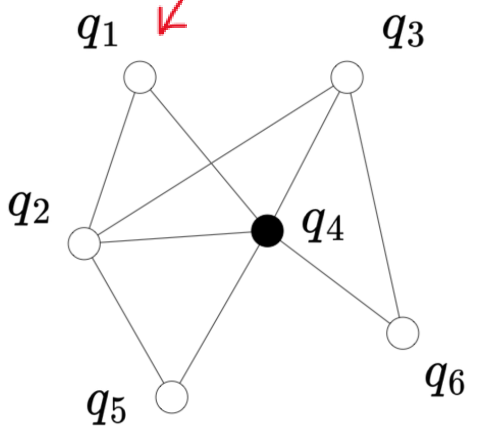
‘*Collective Action Propensity **Score***’

MRU



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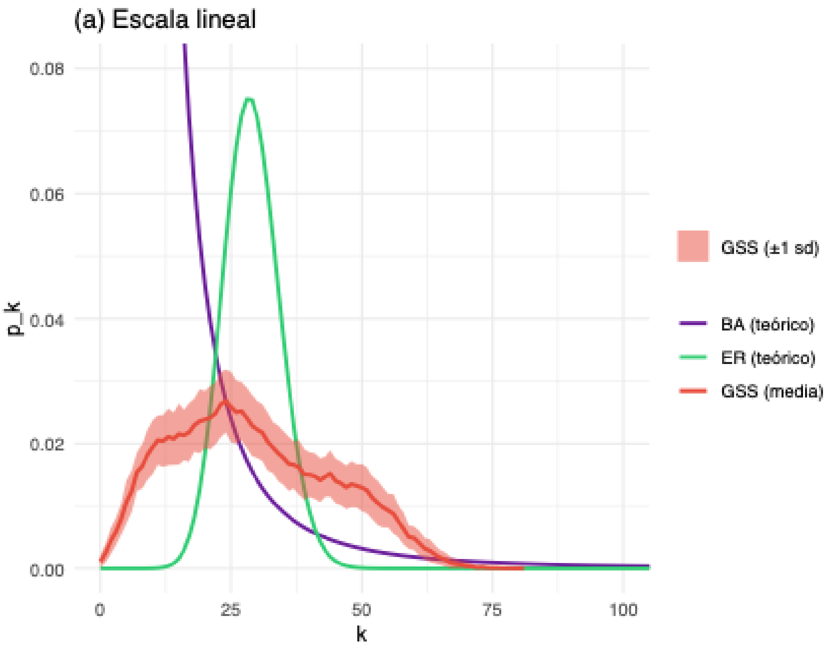

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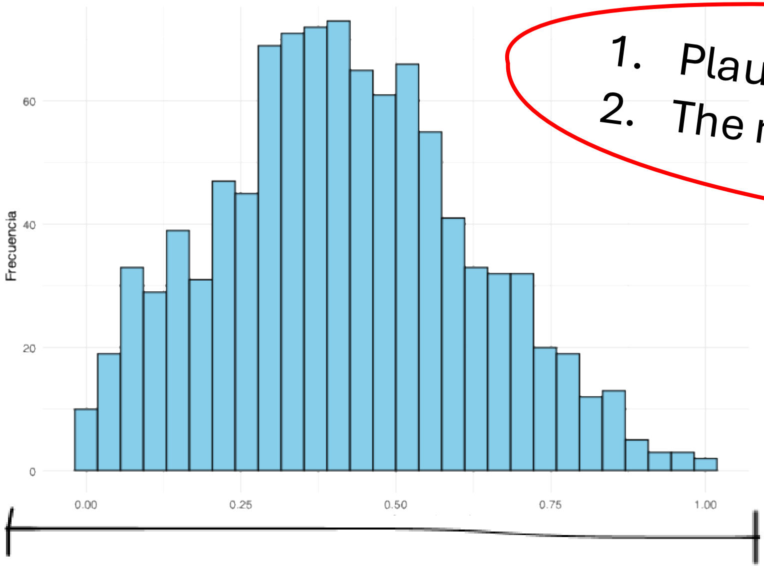

Prone to collective action

0

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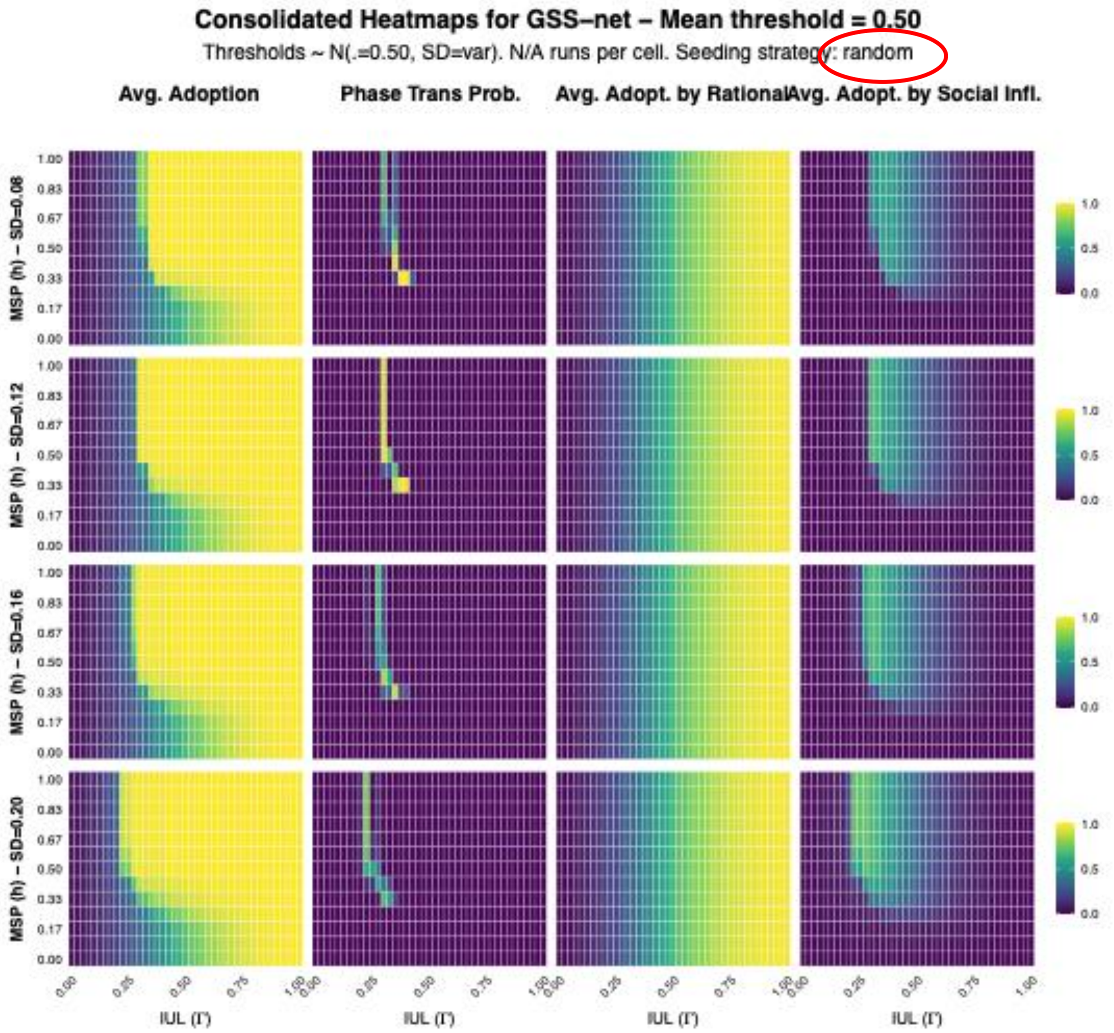
Aversion to collective action



Diffusion of Innovation results remain almost the same for Collective Action!

Setup

Using the ATP-networks (N=1000 nodes):

Parameter Space Exploration:

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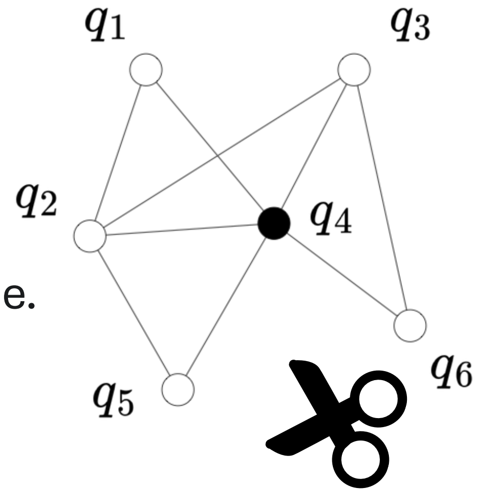
- **Network Structure:**
  Pseudo-empirical ATP/GSS, Erdös-Rényi.



Results – 2nd part

```
Family: binomial
Link function: logit

Formula:
adopt_total ~ s(IUL, MSP, k = 10, by = contagion_type) + tau_mean + tau_sd +
    seed_type + network_type
```

```
Family: binomial
Link function: logit

Formula:
adopt_rational ~ s(IUL, MSP, k = 10, by = contagion_type) + tau_mean + tau_sd +
    seed_type + network_type
```

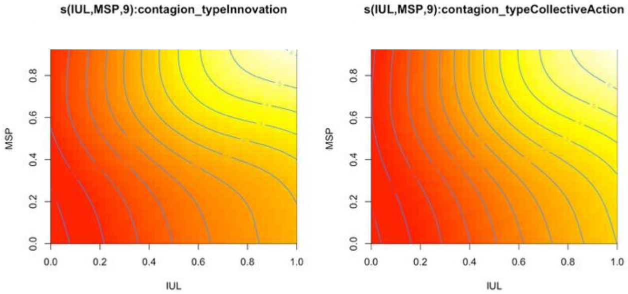
```
Family: binomial
Link function: logit

Formula:
adopt_social ~ s(IUL, MSP, k = 10, by = contagion_type) + tau_mean + tau_sd +
    seed_type + network_type
```

Results – 2nd part

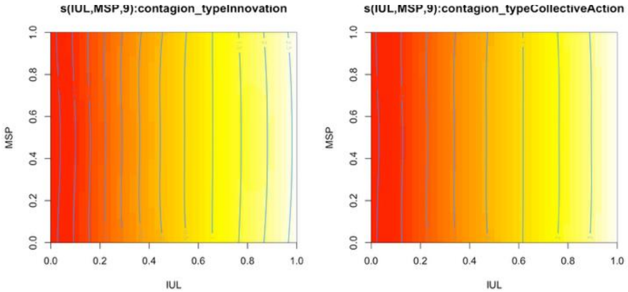
```
Family: binomial
Link function: logit

Formula:
adopt_total ~ s(IUL, MSP, k = 10, by = contagion_type) + tau_mean + tau_sd +
              seed_type + network_type
```



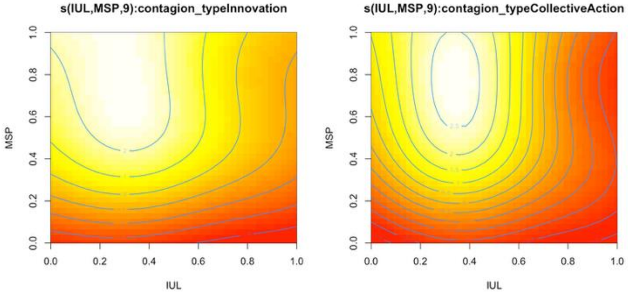

```
Family: binomial
Link function: logit

Formula:
adopt_rational ~ s(IUL, MSP, k = 10, by = contagion_type) + tau_mean + tau_sd +
                 seed_type + network_type
```




```
Family: binomial
Link function: logit

Formula:
adopt_social ~ s(IUL, MSP, k = 10, by = contagion_type) + tau_mean + tau_sd +
               seed_type + network_type
```



Results – 2nd part

```
Family: binomial
Link function: logit

Formula:
adopt_total ~ s(IUL, MSP, k = 10, by = contagion_type) + tau_mean + tau_sd +
              seed_type + network_type

Parametric coefficients:
                     Estimate   OR  Std.Error  z value Pr(>|z|)
(Intercept)           5.304    -   1.997e-04   26557   <2e-16 ***
tau_mean             -6.739    -   2.609e-04  -25829   <2e-16 ***
tau_sd                5.717    -   6.079e-04    9405   <2e-16 ***
seed_typecentral      0.137  1.147 8.527e-05    1610   <2e-16 ***
seed_typecloseness    0.132  1.141 8.525e-05    1548   <2e-16 ***
seed_typeeigen        0.134  1.145 8.526e-05    1581   <2e-16 ***
seed_typemarginal    -0.185  0.831 8.450e-05   -2189   <2e-16 ***
network_typeER       -0.376  0.687 5.411e-05   -6957   <2e-16 ***
---
Approximate significance of smooth terms:
                                        k' edf k-index p-value
s(IUL,MSP):contagion_typeInnovation       9   9    0.95   0.005 **
s(IUL,MSP):contagion_typeCollectiveAction 9   9    0.95  <2e-16 ***
---
R-sq.(adj) =  0.855   Deviance explained = 84.4%
fREML = 8.5075e+08  Scale est. = 1         n = 170560
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Results – 2^{nd} part

```
Family: binomial
Link function: logit

Formula:
adopt_total ~ s(IUL, MSP, k = 10, by = contagion_type) + tau_mean + tau_sd +
              seed_type + network_type

Parametric coefficients:
                     Estimate  OR   Std.Error  z value Pr(>|z|)
(Intercept)           5.304    -    1.997e-04   26557   <2e-16 ***
tau_mean             -6.739    -    2.609e-04  -25829   <2e-16 ***
tau_sd                5.717    -    6.079e-04    9405   <2e-16 ***
seed_typecentral      0.137  1.147 8.527e-05    1610   <2e-16 ***
seed_typecloseness    0.132  1.141 8.525e-05    1548   <2e-16 ***
seed_typeeigen        0.134  1.145 8.526e-05    1581   <2e-16 ***
seed_typemarginal    -0.185  0.831 8.450e-05   -2189   <2e-16 ***
network_typeER       -0.376  0.687 5.411e-05   -6957   <2e-16 ***
---
Approximate significance of smooth terms:
                                         k' edf k-index p-value
s(IUL,MSP):contagion_typeInnovation       9   9    0.95   0.005 **
s(IUL,MSP):contagion_typeCollectiveAction 9   9    0.95  <2e-16 ***
---
R-sq.(adj) =  0.855   Deviance explained = 84.4%
fREML = 8.5075e+08  Scale est. = 1         n = 170560
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
Family: binomial
Link function: logit

Formula:
adopt_rational ~ s(IUL, MSP, k = 10, by = contagion_type) + tau_mean + tau_sd +
                 seed_type + network_type

Parametric coefficients:
                     Estimate  OR   Std.Error  z value Pr(>|z|)
(Intercept)           0.584    -    1.214e-04  4818.2   <2e-16 ***
tau_mean             -0.216    -    1.904e-04 -1138.0   <2e-16 ***
tau_sd                0.058    -    4.759e-04   122.5   <2e-16 ***
seed_typecentral     -0.053  0.948 6.733e-05  -789.0   <2e-16 ***
seed_typecloseness   -0.050  0.950 6.734e-05  -753.2   <2e-16 ***
seed_typeeigen       -0.051  0.950 6.733e-05  -772.0   <2e-16 ***
seed_typemarginal    -0.008  0.991 6.740e-05  -129.2   <2e-16 ***
network_typeER        0.056  1.058 4.257e-05  1317.5   <2e-16 ***
---
Approximate significance of smooth terms:
                                         k' edf k-index p-value
s(IUL,MSP):contagion_typeInnovation       9   9     0.6  <2e-16 ***
s(IUL,MSP):contagion_typeCollectiveAction 9   9     0.6  <2e-16 ***
---
R-sq.(adj) =  0.986   Deviance explained = 98.2%
fREML = 7.8107e+07  Scale est. = 1         n = 170560
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Results – 2nd part

```
Family: binomial
Link function: logit

Formula:
adopt_total ~ s(IUL, MSP, k = 10, by = contagion_type) + tau_mean + tau_sd +
    seed_type + network_type

Parametric coefficients:
                     Estimate   OR  Std.Error  z value Pr(>|z|)
(Intercept)             5.304    -  1.997e-04   26557   <2e-16 ***
tau_mean               -6.739    -  2.609e-04  -25829   <2e-16 ***
tau_sd                  5.717    -  6.079e-04    9405   <2e-16 ***
seed_typecentral        0.137 1.147 8.527e-05    1610   <2e-16 ***
seed_typecloseness      0.132 1.141 8.525e-05    1548   <2e-16 ***
seed_typeeigen          0.134 1.145 8.526e-05    1581   <2e-16 ***
seed_typemarginal      -0.185 0.831 8.450e-05   -2189   <2e-16 ***
network_typeER         -0.376 0.687 5.411e-05   -6957   <2e-16 ***
---
Approximate significance of smooth terms:
                                         k' edf k-index p-value
s(IUL,MSP):contagion_typeInnovation       9   9    0.95   0.005 **
s(IUL,MSP):contagion_typeCollectiveAction 9   9    0.95  <2e-16 ***
---
R-sq.(adj) =  0.855   Deviance explained = 84.4%
fREML = 8.5075e+08  Scale est. = 1         n = 170560
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

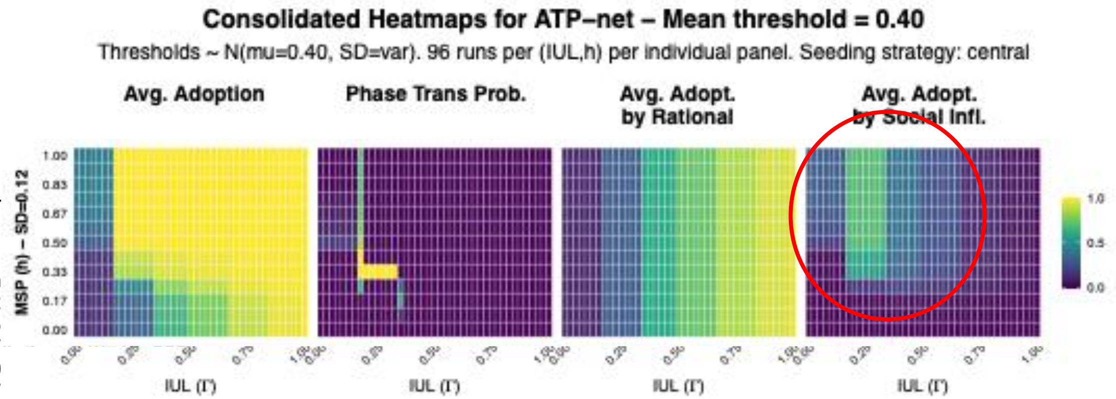
```
Family: binomial
Link function: logit

Formula:
adopt_social ~ s(IUL, MSP, k = 10, by = contagion_type) + tau_mean + tau_sd +
    seed_type + network_type

Parametric coefficients:
                     Estimate   OR  Std.Error  z value Pr(>|z|)
(Intercept)            -0.543    -  1.422e-04  -3822.5   <2e-16 ***
tau_mean               -5.784    -  2.337e-04 -24751.3   <2e-16 ***
tau_sd                  4.703    -  5.550e-04   8474.5   <2e-16 ***
seed_typecentral        0.047 1.048 7.781e-05    608.9   <2e-16 ***
seed_typecloseness      0.045 1.046 7.783e-05    579.1   <2e-16 ***
seed_typeeigen          0.047 1.048 7.781e-05    607.8   <2e-16 ***
seed_typemarginal      -0.080 0.922 7.874e-05  -1027.2   <2e-16 ***
network_typeER         -0.340 0.711 4.951e-05  -6886.1   <2e-16 ***
---
Approximate significance of smooth terms:
                                         k' edf k-index p-value
s(IUL,MSP):contagion_typeInnovation       9   9    0.92  <2e-16 ***
s(IUL,MSP):contagion_typeCollectiveAction 9   9    0.92  <2e-16 ***
---
R-sq.(adj) =   0.67   Deviance explained =   69%
fREML = 9.2894e+08  Scale est. = 1         n = 170560
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Conclusions – 2nd part

Conclusions – 2nd part

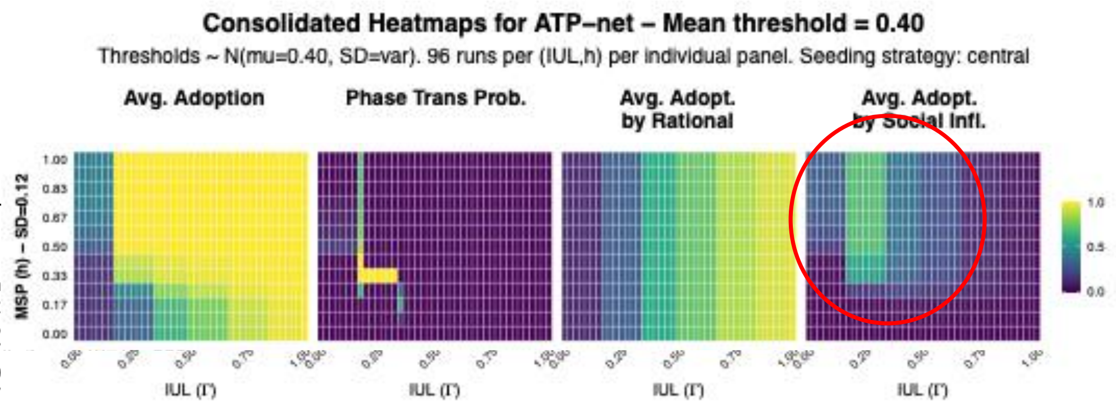



```
Family: binomial
Link function: logit

Formula:
adopt_social ~ s(IUL, MSP, k = 10, by = contagion_
              seed_type + network_type

Parametric coefficients:
                     Estimate  OR   Std.Error  z va
(Intercept)          -0.543    -    1.422e-04  -382
tau_mean             -5.784    -    2.337e-04 -2475
tau_sd                4.703    -    5.550e-04   847
seed_typecentral      0.047  1.048 7.781e-05    60
seed_typecloseness    0.045  1.046 7.783e-05    57
seed_typeeigen        0.047  1.048 7.781e-05   607.8   <2e-16 ***
seed_typemarginal    -0.080  0.922 7.874e-05  -1027.2  <2e-16 ***
network_typeER       -0.340  0.711 4.951e-05  -6886.1  <2e-16 ***
---
Approximate significance of smooth terms:
                                          k' edf k-index p-value
s(IUL,MSP):contagion_typeInnovation        9   9    0.92  <2e-16 ***
s(IUL,MSP):contagion_typeCollectiveAction  9   9    0.92  <2e-16 ***
---
R-sq.(adj) =   0.67   Deviance explained =   69%
fREML = 9.2894e+08  Scale est. = 1         n = 170560
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Conclusions – 2nd part




```
Family: binomial
Link function: logit

Formula:
adopt_social ~ s(IUL, MSP, k = 10, by = contagion_
            seed_type + network_type

Parametric coefficients:
                     Estimate  OR   Std.Error  z va
(Intercept)           -0.543   -    1.422e-04  -382
tau_mean              -5.784   -    2.337e-04 -2475
tau_sd                 4.703   -    5.550e-04   847
seed_typecentral       0.047 1.048 7.781e-05    60
seed_typecloseness     0.045 1.046 7.783e-05    57
seed_typeeigen         0.047 1.048 7.781e-05   607.8    <2e-16 ***
seed_typemarginal     -0.080 0.922 7.874e-05  -1027.2   <2e-16 ***
network_typeER        -0.340 0.711 4.951e-05  -6886.1   <2e-16 ***
---
Approximate significance of smooth terms:
                                    k' edf k-index p-value
s(IUL,MSP):contagion_typeInnovation      9   9    0.92  <2e-16 ***
s(IUL,MSP):contagion_typeCollectiveAction 9  9    0.92  <2e-16 ***
---
R-sq.(adj) =   0.67   Deviance explained =   69%
fREML = 9.2894e+08  Scale est. = 1         n = 170560
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

It seems like
networks decide for us!

Thanks!

All the results are available on the GitHub repository !

References

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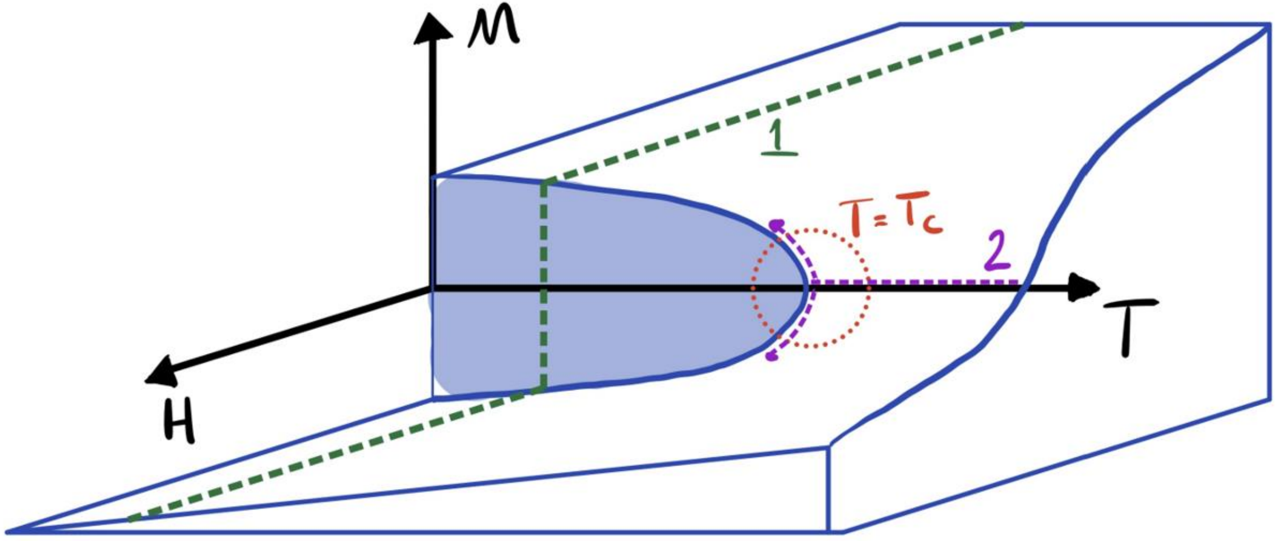
Phase Transitions



Extras

Imputing network structure to a Survey

Table 1. Case-Control Logistic Regression Predicting a Confiding Relationship based on General Social Survey Ego Network Data, 1985 and 2004.

Variable	Model 1
Intercept	−14.456***
	(0.048)
Different race	−1.819***
	(0.077)
Different religion	−1.362***
	(0.044)
Different sex	−0.317***
	(0.025)
Education difference	−0.049***
	(0.002)
Age difference	−0.173***
	(0.009)
Year	−0.179***
	(0.047)
N (respondents)	3,001
N (dyads)	1,139,161

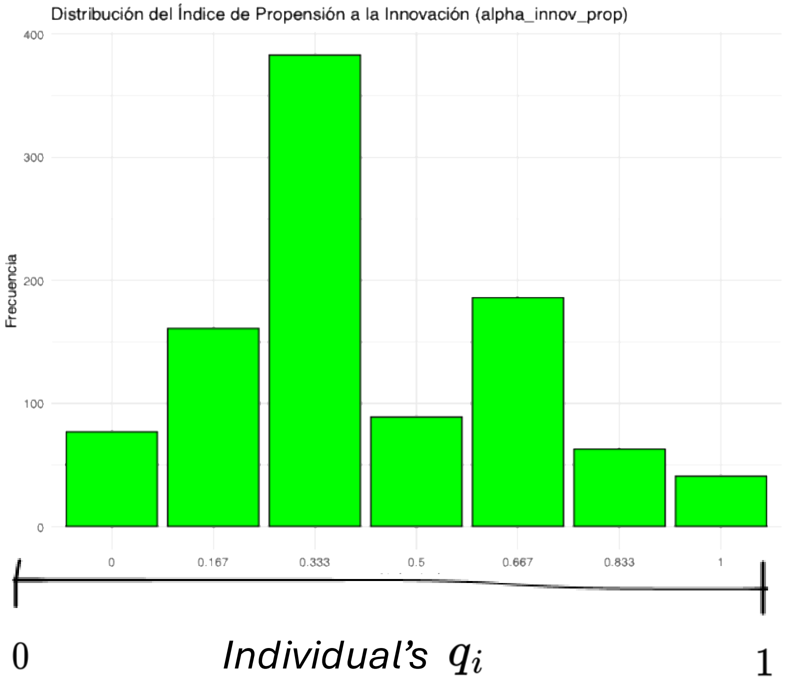
Imputing network structure to a Survey

Items to create the '*Openness to Innovations Score*' from ATP 2014 **(binary):**

1. Usually try new products before others do. → pro-innov.
2. Prefer my tried and trusted brands. → **anti**-innov.
3. Like being able to tell others about new brands and products I have tried. → pro-innov.
4. Like the variety of trying new products. → pro-innov.
5. Feel more comfortable using familiar brands and products. → **anti**-innov.
6. Wait until I hear about others' experiences before I try new products. → **anti**-innov.

Imputing network structure to a Survey

Items to create the '*Openness to Innovations Score*' from ATP 2014 **(binary):**



Prone to innovations

Individual's q_i

Aversion to innovation

Imputing network structure to a Survey

Here are some forms of political and social action that people can take.
Please indicate, for *each one*,

(0) whether you have done any of these things in the **past year**,

(1) have done it in the **more distant past**,

(2) have not done it but **might do it**,

(3) or have not done it and **would never**, under any circumstances, do it.

1. Signed a petition
2. Boycotted, or deliberately bought, certain products for political, ethical or environmental reasons.
3. Took part in a demonstration.
4. Attended a political meeting or rally.
5. Contacted, or attempted to contact, a politician or a civil servant to express your views.
6. Donated money or raised funds for a social or political activity.

Imputing network structure to a Survey

Here are some forms of political and social action that people can take.
Please indicate, for *each one*,

(0) whether you have done any of these things in the **past year**,
(1) have done it in the **more distant past**,
(2) have not done it but **might do it**,
(3) or have not done it and **would never**, under any circumstances, do it.

7. Contacted or appeared in the media to express your views.
8. Joined an Internet political forum or discussion group.

9. Suppose a law were being considered by the Congress that you considered to be unjust or harmful. If such a case arose, how likely is it that you, acting along or together with others, would be able to try to do something about it?

0) Very likely 1) Fairly likely 2) Not very likely 3) Not at all likely